

CSYE 6225: Network Structure & Cloud Computing Course Assignment # 2

1. Basic Network Information

```
[(base) zianw@wangziyandeMacBook-Pro ~ % ip addr show ]
1: lo0: <UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384 status UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8
    inet6 ::1/128
    inet6 fe80::1/64
2: gif0: <POINTOPOINT,MULTICAST> mtu 1280 status UNKNOWN
    link/none
3: stf0: <> mtu 1280 status UNKNOWN
    link/unknown
4: anp1: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:12 brd ff:ff:ff:ff:ff:ff
5: anp10: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:11 brd ff:ff:ff:ff:ff:ff
6: en7: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:f1 brd ff:ff:ff:ff:ff:ff
7: en8: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:f2 brd ff:ff:ff:ff:ff:ff
8: en1: <UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:80 brd ff:ff:ff:ff:ff:ff
9: en2: <UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:84 brd ff:ff:ff:ff:ff:ff
10: bridge0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:80 brd ff:ff:ff:ff:ff:ff
11: ap1: <UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 3e:06:30:13:38:2a brd ff:ff:ff:ff:ff:ff
    inet6 fe80::3c06:30ff:fe13:382a/64
12: en0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status UP
    link/ether 3c:06:30:13:38:2a brd ff:ff:ff:ff:ff:ff
    inet6 fe80::87:4f40:2395:f3b3/64
    inet6 2601:646:a180:5db0:1c24:59c4:511d:40b/64
    inet6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34/64
    inet 10.0.0.13/24 brd 10.0.0.255
    inet6 2601:646:a180:5db0::3c30/64
13: awd10: <UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status UP
    link/ether 82:e8:6f:75:7f:88 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::80e8:6fff:fe75:7f88/64
14: llw0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 82:e8:6f:75:7f:88 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::80e8:6fff:fe75:7f88/64
15: utun0: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500 status UNKNOWN
    link/none
    inet6 fe80::c5a0:fc3a:bb65:12d/64
16: utun1: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380 status UNKNOWN
    link/none
    inet6 fe80::6c50:3722:7ce:dbec/64
17: utun2: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000 status UNKNOWN
    link/none
    inet6 fe80::af09:3c6b:3f63:bf84/64
18: utun3: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000 status UNKNOWN
    link/none
    inet6 fe80::ce81:b1c:bd2c:69e/64
```

```
[(base) ziyaw@wangziyandeMacBook-Pro ~ % ip link show
1: lo0: <UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384 status UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: gif0: <POINTOPOINT,MULTICAST> mtu 1280 status UNKNOWN
    link/none
3: stf0: <> mtu 1280 status UNKNOWN
    link/unknown
4: anpi1: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:12 brd ff:ff:ff:ff:ff:ff
5: anpi0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:11 brd ff:ff:ff:ff:ff:ff
6: en7: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:f1 brd ff:ff:ff:ff:ff:ff
7: en8: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether ba:27:35:c4:da:f2 brd ff:ff:ff:ff:ff:ff
8: en1: <UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:80 brd ff:ff:ff:ff:ff:ff
9: en2: <UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:84 brd ff:ff:ff:ff:ff:ff
10: bridge0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 36:2a:90:4a:df:80 brd ff:ff:ff:ff:ff:ff
11: ap1: <UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 3e:06:30:13:38:2a brd ff:ff:ff:ff:ff:ff
12: en0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status UP
    link/ether 3c:06:30:13:38:2a brd ff:ff:ff:ff:ff:ff
13: awdl0: <UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status UP
    link/ether 82:e8:6f:75:7f:88 brd ff:ff:ff:ff:ff:ff
14: llw0: <UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500 status DOWN
    link/ether 82:e8:6f:75:7f:88 brd ff:ff:ff:ff:ff:ff
15: utun0: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500 status UNKNOWN
    link/none
16: utun1: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380 status UNKNOWN
    link/none
17: utun2: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000 status UNKNOWN
    link/none
18: utun3: <UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000 status UNKNOWN
    link/none
[(base) ziyaw@wangziyandeMacBook-Pro ~ % ip route show
default via 10.0.0.1 dev en0
10.0.0.0/24 dev en0 scope link
10.0.0.1/32 dev en0 scope link
10.0.0.13/32 dev en0 scope link
127.0.0.0/8 via 127.0.0.1 dev lo0
127.0.0.1/32 via 127.0.0.1 dev lo0
169.254.0.0/16 dev en0 scope link
224.0.0.0/4 dev en0 scope link
255.255.255.255/32 dev en0 scope link
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % cat /etc/resolv.conf
#
# macOS Notice
#
# This file is not consulted for DNS hostname resolution, address
# resolution, or the DNS query routing mechanism used by most
# processes on this system.
#
# To view the DNS configuration used by this system, use:
#   scutil --dns
#
# SEE ALSO
#   dns-sd(1), scutil(8)
#
# This file is automatically generated.
#
search hsd1.ca.comcast.net
nameserver 2001:558:feed::1
nameserver 2001:558:feed::2
nameserver 75.75.75.75
nameserver 75.75.76.76
```

2. Network Statistics

```
(base) zianw@wangziyandeMacBook-Pro ~ % netstat -i
```

Name	Mtu	Network	Address	Ipkts	Ierrs	Opkts	Oerrs	Coll
lo0	16384	<Link#1>		536109	0	536109	0	0
lo0	16384	127	localhost	536109	-	536109	-	-
lo0	16384	localhost	::1	536109	-	536109	-	-
lo0	16384	wangziyande	fe80:1::1	536109	-	536109	-	-
gif0*	1280	<Link#2>		0	0	0	0	0
stf0*	1280	<Link#3>		0	0	0	0	0
anpi1	1500	<Link#4>	ba:27:35:c4:da:12	0	0	0	0	0
anpi0	1500	<Link#5>	ba:27:35:c4:da:11	0	0	0	0	0
en7	1500	<Link#6>	ba:27:35:c4:da:f1	0	0	0	0	0
en8	1500	<Link#7>	ba:27:35:c4:da:f2	0	0	0	0	0
en1	1500	<Link#8>	36:2a:90:4a:df:80	0	0	0	0	0
en2	1500	<Link#9>	36:2a:90:4a:df:84	0	0	0	0	0
bridge0	1500	<Link#10>	36:2a:90:4a:df:80	0	0	0	0	0
ap1	1500	<Link#11>	3e:06:30:13:38:2a	0	0	62857	0	0
ap1	1500	wangziyande	fe80:b::3c06:30ff	0	-	62857	-	-
en0	1500	<Link#12>	3c:06:30:13:38:2a	55483245	0	15996749	0	0
en0	1500	wangziyande	fe80:c::87:4f40:2	55483245	-	15996749	-	-
en0	1500	2601:646:a1	2601:646:a180:5db	55483245	-	15996749	-	-
en0	1500	2601:646:a1	2601:646:a180:5db	55483245	-	15996749	-	-
en0	1500	10/24	10.0.0.13	55483245	-	15996749	-	-
en0	1500	2601:646:a1	2601:646:a180:5db	55483245	-	15996749	-	-
awdl0	1500	<Link#13>	82:e8:6f:75:7f:88	2723	0	2917	0	0
awdl0	1500	fe80::80e8:	fe80:d::80e8:6fff	2723	-	2917	-	-
llw0	1500	<Link#14>	82:e8:6f:75:7f:88	0	0	0	0	0
llw0	1500	fe80::80e8:	fe80:e::80e8:6fff	0	-	0	-	-
utun0	1500	<Link#15>		0	0	1	0	0
utun0	1500	wangziyande	fe80:f::c5a0:fc3a	0	-	1	-	-
utun1	1380	<Link#16>		0	0	2082	0	0
utun1	1380	wangziyande	fe80:10::6c50:372	0	-	2082	-	-
utun2	2000	<Link#17>		0	0	2082	0	0
utun2	2000	wangziyande	fe80:11::af09:3c6	0	-	2082	-	-
utun3	1000	<Link#18>		0	0	2082	0	0
utun3	1000	wangziyande	fe80:12::ce81:b1c	0	-	2082	-	-

```
(base) zianw@wangziyandeMacBook-Pro ~ % netstat -s
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % netstat -s
tcp:
```

```
0 packet sent
    0 data packet (0 byte)
    0 data packet (0 byte) retransmitted
    0 resend initiated by MTU discovery
    0 ack-only packet (0 delayed)
    0 URG only packet
    0 window probe packet
    0 window update packet
    0 control packet
    0 data packet sent after flow control
    0 challenge ACK sent due to unexpected SYN
    0 challenge ACK sent due to unexpected RST
    0 checksummed in software
        0 segment (0 byte) over IPv4
        0 segment (0 byte) over IPv6
0 packet received
    0 ack (for 0 byte)
    0 duplicate ack
    0 ack for unsent data
    0 packet (0 byte) received in-sequence
    0 completely duplicate packet (0 byte)
    0 old duplicate packet
    0 received packet dropped due to low memory
    0 packet with some dup. data (0 byte duped)
    0 out-of-order packet (0 byte)
    0 packet (0 byte) of data after window
    0 window probe
    0 window update packet
    0 packet recovered after loss
    0 packet received after close
    0 bad reset
    0 discarded for bad checksum
    0 checksummed in software
        0 segment (0 byte) over IPv4
        0 segment (0 byte) over IPv6
    0 discarded for bad header offset field
    0 discarded because packet too short
0 connection request
0 connection accept
0 bad connection attempt
0 listen queue overflow
0 connection established (including accepts)
0 connection closed (including 0 drop)
    0 connection updated cached RTT on close
    0 connection updated cached RTT variance on close
    0 connection updated cached ssthresh on close
    0 connection initialized RTT from route cache
    0 connection initialized RTT variance from route cache
    0 connection initialized ssthresh from route cache
0 embryonic connection dropped
0 segment updated rtt (of 0 attempt)
0 retransmit timeout
    0 connection dropped by rexmit timeout
    0 connection dropped after retransmitting FIN
```



```

117938 IPv4 stream sockets created since boot
4660212 IPv4 datagram sockets created since boot
439491 IPv4 datagram sockets connected
460366 IPv4 DNS sockets
4529528 IPv4 datagram sockets without data
641857 IPv6 stream sockets created since boot
2299138 IPv6 datagram sockets created since boot
776677 IPv6 datagram sockets connected
0 IPv6 DNS socket
1830424 IPv6 datagram sockets without data
21081 socket multicast joins since boot
21081 socket multicast joins since boot by OS
37384 IPv4 stream nexus flows added since boot
10144 IPv4 datagram nexus flows added since boot
27471 IPv6 stream nexus flows added since boot
1864 IPv6 datagram nexus flows added since boot
18 interfaces currently allocated
18 interfaces allocated since boot
18 interfaces currently allocated by OS
18 extended interfaces allocated since boot by OS
0 PF addrule operation since boot
0 PF addrule operation since boot by OS
0 vmnet start since boot
if_ports_used:
123 wakeuuid generations
0 offload port list query with wakeuuid not set
34934 total offload port entries created since boot
376 current offload port entries
2530 max offload port entries
75078 duplicate offload port entries
1473260 total table entry searches
100 max hash table entry searches
12 match so wake packet calls
16 match ch wake packet calls
28 IPv4 wake packets
0 IPv6 wake packet
28 TCP wake packets
0 UDP wake packet
0 ISAKMP NAT traversal wake packet
0 ESP wake packet
0 bad protocol wake packet
0 bad family wake packet
11 wake packet events
17 duplicate wake packet events in same wake cycle
0 wake packet event undelivered
0 unattributed wake packet event
0 duplicate unattributed wake packet event in same wake cycle
0 unattributed wake packet event undelivered
0 unattributed wake packet received with null interface
0 bad packet without wake flag
0 pure fragment wake packet
0 packet with incomplete TCP header
0 packet with incomplete UDP header
0 port entry not added with wakeuuid not set
0 deferred matching of ISAKMP NAT traversal wake packet
0 spurious wake packet event

```

```

[(base) zianw@wangziyandeMacBook-Pro ~ % ifstat
zsh: command not found: ifstat
(base) zianw@wangziyandeMacBook-Pro ~ % brew install ifstat

==> Downloading https://formulae.brew.sh/api/formula.jws.json
==> Downloading https://formulae.brew.sh/api/cask.jws.json
==> Downloading https://ghcr.io/v2/homebrew/core/ifstat/manifests/1.1
##### 100.0%
==> Fetching ifstat
==> Downloading https://ghcr.io/v2/homebrew/core/ifstat/blobs/sha256:077e591c44a830e200e6587ea68eef
##### 100.0%
==> Pouring ifstat--1.1.arm64_sonoma.bottle.tar.gz
📦 /opt/homebrew/Cellar/ifstat/1.1: 9 files, 92.4KB
==> Running `brew cleanup ifstat`...
Disable this behaviour by setting HOMEBREW_NO_INSTALL_CLEANUP.
Hide these hints with HOMEBREW_NO_ENV_HINTS (see `man brew`).
(base) zianw@wangziyandeMacBook-Pro ~ % which ifstat

/opt/homebrew/bin/ifstat
[(base) zianw@wangziyandeMacBook-Pro ~ % ifstat

```

	anpi1		anpi0		en7		en8		en1	
	en2	llw0	bridge0	utun0	ap1	utun1	en0	utun2	awdl0	utun3
KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in	KB/s in
KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out	KB/s out
t	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.95	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	8.06	5.26	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	6.81	5.41	0.00	0.00
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

3. Packet Analysis

```
(base) zianw@wangziyandeMacBook-Pro ~ % sudo tcpdump -i en0
```

[Password:

```
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
10:47:33.796825 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:47:33.900272 IP6 2602:f79a::1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51783: Flags [P.],
seq 1688379035:1688379089, ack 2647020107, win 10, options [nop,nop,TS val 275181400 ecr 2717622869
], length 54
10:47:33.900623 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51783 > 2602:f79a::1.https: Flags [., a
ck 54, win 2048, options [nop,nop,TS val 2717629217 ecr 275181400], length 0
10:47:33.901721 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51783 > 2602:f79a::1.https: Flags [P.],
seq 1:59, ack 54, win 2048, options [nop,nop,TS val 2717629219 ecr 275181400], length 58
10:47:33.923346 IP6 2602:f79a::1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51783: Flags [., a
ck 59, win 10, options [nop,nop,TS val 275181496 ecr 2717629219], length 0
10:47:33.954163 IP 10.0.0.13.mdns > mdns.mcast.net.mdns: 0 PTR (QU)? 2.9.7.8.c.c.e.f.f.6.1.f.e.a.
8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:33.954217 IP6 wangziyandemacbook-pro.local.mdns > ff02::fb.mdns: 0 PTR (QU)? 2.9.7.8.c.c.e.f.
f.f.6.1.f.e.a.8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:34.718384 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:47:34.957926 IP 10.0.0.13.mdns > mdns.mcast.net.mdns: 0 PTR (QM)? 2.9.7.8.c.c.e.f.f.6.1.f.e.a.
8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:34.957987 IP6 wangziyandemacbook-pro.local.mdns > ff02::fb.mdns: 0 PTR (QM)? 2.9.7.8.c.c.e.f.
f.f.6.1.f.e.a.8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:35.707391 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.50845 > 2001:558:feed:443::183.https: Fl
ags [., ack 3060802292, win 2048, length 0
10:47:35.743969 IP6 2001:558:feed:443::183.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.50845: Fl
ags [., ack 1, win 51120, options [nop,nop,TS val 2017191339 ecr 3641183509], length 0
10:47:35.890399 IP6 2600:1f18:24e6:b902:a46c:a4a6:87fe:c14c.https > 2601:646:a180:5db0:4df4:e8f3:da
b4:2e34.51906: Flags [P.], seq 3882334983:3882335022, ack 2951999263, win 81, options [nop,nop,TS v
al 4074756226 ecr 1956155131], length 39
10:47:35.890552 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51906 > 2600:1f18:24e6:b902:a46c:a4a6:87
fe:c14c.https: Flags [., ack 39, win 2047, options [nop,nop,TS val 1956215121 ecr 4074756226], len
gth 0
10:47:35.890835 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51906 > 2600:1f18:24e6:b902:a46c:a4a6:87
fe:c14c.https: Flags [F.], seq 1, ack 39, win 2048, options [nop,nop,TS val 1956215121 ecr 40747562
26], length 0
10:47:35.942134 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:47:35.979794 IP6 2600:1f18:24e6:b902:a46c:a4a6:87fe:c14c.https > 2601:646:a180:5db0:4df4:e8f3:da
b4:2e34.51906: Flags [P.], seq 39:63, ack 2, win 81, options [nop,nop,TS val 4074756315 ecr 1956215
121], length 24
10:47:35.979797 IP6 2600:1f18:24e6:b902:a46c:a4a6:87fe:c14c.https > 2601:646:a180:5db0:4df4:e8f3:da
b4:2e34.51906: Flags [F.], seq 63, ack 2, win 81, options [nop,nop,TS val 4074756315 ecr 1956215121
], length 0
10:47:35.980045 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51906 > 2600:1f18:24e6:b902:a46c:a4a6:87
fe:c14c.https: Flags [R], seq 2951999264, win 0, length 0
10:47:35.980119 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51906 > 2600:1f18:24e6:b902:a46c:a4a6:87
fe:c14c.https: Flags [R], seq 2951999264, win 0, length 0
10:47:36.865883 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:47:36.865885 IP 10.0.0.142.62525 > broadcasthost.6667: UDP, length 204
10:47:37.791830 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:47:37.964154 IP 10.0.0.13.mdns > mdns.mcast.net.mdns: 0 PTR (QM)? 2.9.7.8.c.c.e.f.f.6.1.f.e.a.
8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:37.964224 IP6 wangziyandemacbook-pro.local.mdns > ff02::fb.mdns: 0 PTR (QM)? 2.9.7.8.c.c.e.f.
f.f.6.1.f.e.a.8.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
10:47:38.712058 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
```

```

Flags [.], ack 17017, win 2048, options [nop,nop,TS val 661257452 ecr 515211478], length 0
10:48:05.747021 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
10:48:05.849429 IP6 2606:4700:4400::ac40:9bd1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862:
Flags [P.], seq 17017:17085, ack 47917, win 22, options [nop,nop,TS val 515211850 ecr 661257452],
length 68
10:48:05.849778 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862 > 2606:4700:4400::ac40:9bd1.https:
Flags [.], ack 17085, win 2048, options [nop,nop,TS val 661257901 ecr 515211850], length 0
10:48:05.851515 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862 > 2606:4700:4400::ac40:9bd1.https:
Flags [P.], seq 47917:47952, ack 17085, win 2048, options [nop,nop,TS val 661257902 ecr 515211850]
, length 35
10:48:05.874274 IP6 2606:4700:4400::ac40:9bd1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862:
Flags [.], ack 47952, win 22, options [nop,nop,TS val 515211952 ecr 661257902], length 0
10:48:05.966741 IP6 2606:4700:4400::ac40:9bd1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862:
Flags [P.], seq 17085:17169, ack 47952, win 22, options [nop,nop,TS val 515212043 ecr 661257902],
length 84
10:48:05.966963 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51862 > 2606:4700:4400::ac40:9bd1.https:
Flags [.], ack 17169, win 2048, options [nop,nop,TS val 661258018 ecr 515212043], length 0
^C
512 packets captured
512 packets received by filter
0 packets dropped by kernel

```



```
sudo tcpdump -i en0 port 80
```

```
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
11:29:24.385317 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [S], seq 2493148969, win 65535, options [mss 1460,nop,wscale 6,nop,nop,TS val 3285764368 ecr 0,sackOK,eol], length 0
11:29:24.407035 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [S.], seq 3851829278, ack 2493148970, win 65160, options [mss 1424,sackOK,TS val 1243467089 ecr 3285764368,nop,wscale 7], length 0
11:29:24.407157 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [.], ack 1, win 2051, options [nop,nop,TS val 3285764390 ecr 1243467089], length 0
11:29:24.407427 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [P.], seq 1:980, ack 1, win 2051, options [nop,nop,TS val 3285764390 ecr 1243467089], length 979: HTTP: POST /mmtls/273efb30 HTTP/1.1
11:29:24.437307 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [.], ack 980, win 502, options [nop,nop,TS val 1243467117 ecr 3285764390], length 0
11:29:24.693982 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [P.], seq 1:1293, ack 980, win 502, options [nop,nop,TS val 1243467371 ecr 3285764390], length 1292: HTTP: HTTP/1.1 200 OK
11:29:24.693984 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [.], seq 1293:2705, ack 980, win 502, options [nop,nop,TS val 1243467371 ecr 3285764390], length 1412: HTTP
11:29:24.693985 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [P.], seq 2705:2943, ack 980, win 502, options [nop,nop,TS val 1243467371 ecr 3285764390], length 238: HTTP
11:29:24.693986 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [F.], seq 2943, ack 980, win 502, options [nop,nop,TS val 1243467373 ecr 3285764390], length 0
11:29:24.694138 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [.], ack 2943, win 2005, options [nop,nop,TS val 3285764677 ecr 1243467371], length 0
11:29:24.694201 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [.], ack 2944, win 2005, options [nop,nop,TS val 3285764677 ecr 1243467373], length 0
11:29:24.694927 IP 10.0.0.13.52037 > 43.130.30.185.http: Flags [F.], seq 980, ack 2944, win 2048, options [nop,nop,TS val 3285764677 ecr 1243467373], length 0
11:29:24.715025 IP 43.130.30.185.http > 10.0.0.13.52037: Flags [.], ack 981, win 502, options [nop,nop,TS val 1243467397 ecr 3285764677], length 0
11:29:36.179640 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [S], seq 2340254549, win 65535, options [mss 1460,nop,wscale 6,nop,nop,TS val 882801481 ecr 0,sackOK,eol], length 0
11:29:36.215146 IP 43.130.30.185.http > 10.0.0.13.52038: Flags [S.], seq 230084935, ack 2340254550, win 64240, options [mss 1424,nop,nop,sackOK,nop,wscale 7], length 0
11:29:36.215221 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [.], ack 1, win 4096, length 0
11:29:36.215415 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [P.], seq 1:739, ack 1, win 4096, length 738: HTTP: POST /mmtls/27420f04 HTTP/1.1
11:29:36.239579 IP 43.130.30.185.http > 10.0.0.13.52038: Flags [.], ack 739, win 501, length 0
11:29:36.406849 IP 43.130.30.185.http > 10.0.0.13.52038: Flags [P.], seq 1:332, ack 739, win 501, length 331: HTTP: HTTP/1.1 200 OK
11:29:36.406929 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [.], ack 332, win 4090, length 0
11:29:36.407441 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [F.], seq 739, ack 332, win 4096, length 0
11:29:36.431033 IP 43.130.30.185.http > 10.0.0.13.52038: Flags [F.], seq 332, ack 740, win 501, length 0
11:29:36.431145 IP 10.0.0.13.52038 > 43.130.30.185.http: Flags [.], ack 333, win 4096, length 0
^C
23 packets captured
406 packets received by filter
0 packets dropped by kernel
```

iftop

```
interface: en0
IP address is: 10.0.0.13
MAC address is: 3c:06:30:13:38:2a
```

	1.91Mb	3.81Mb	5.72Mb	7.63Mb	9.54Mb			
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2606:4700:4400::ac40:9bd1		0b	12.7Kb	29.4Kb		
	<=			0b	1.49Kb	7.01Kb		
10.0.0.13	=>	lcausi-in-f94.1e100.net		0b	3.91Kb	0.98Kb		
	<=			0b	6.52Kb	1.63Kb		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2602:f79a::2		0b	4.02Kb	1.00Kb		
	<=			0b	4.21Kb	1.05Kb		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2600:1901:0:47fc::		0b	4.58Kb	1.14Kb		
	<=			0b	3.60Kb	923b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2001:558:feed:443::148		0b	2.92Kb	1.15Kb		
	<=			0b	2.74Kb	1.12Kb		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2606:4700:4400::ac40:9bd1		0b	1.47Kb	7.47Kb		
	<=			0b	738b	1.10Kb		
239.255.255.250	=>	169.254.100.1		0b	0b	0b		
	<=			0b	1.39Kb	889b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	cdns01.comcast.net		552b	450b	293b		
	<=			936b	721b	459b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2602:f79a::2		692b	300b	179b		
	<=			964b	351b	766b		
broadcasthost	=>	10.0.0.142		0b	0b	0b		
	<=			0b	371b	371b		
mdns.mcast.net	=>	10.0.0.37		0b	0b	0b		
	<=			0b	242b	106b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2606:4700:4400::ac40:9bd1		0b	0b	54.7Kb		
	<=			0b	0b	54.6Kb		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	sfo03s33-in-x03.1e100.net		0b	0b	747b		
	<=			0b	0b	0.99Kb		
10.0.0.13	=>	17.57.144.119		0b	0b	952b		
	<=			0b	0b	21b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	ussjc2-edge-bx-027.aapling.com		0b	0b	511b		
	<=			0b	0b	367b		
10.0.0.13	=>	ec2-54-147-148-46.compute-1.amazon		0b	0b	595b		
	<=			0b	0b	114b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	2600:1f18:24e6:b901:72e3:1bfe:b271		0b	0b	65b		
	<=			0b	0b	70b		
broadcasthost	=>	0.0.0.0		0b	0b	0b		
	<=			0b	0b	132b		
10.0.0.13	=>	10.0.0.37		0b	0b	88b		
	<=			0b	0b	0b		
10.0.0.13	=>	mdns.mcast.net		0b	0b	30b		
	<=			0b	0b	0b		
2601:646:a180:5db0:4df4:e8f3:dab4:	=>	dz-in-f188.1e100.net		0b	0b	12b		
	<=			0b	0b	14b		
10.0.0.13	=>	ec2-54-83-37-232.compute-1.amazona		0b	0b	8b		
	<=			0b	0b	10b		
1.0.0.10	=>	all-systems.mcast.net		0b	0b	7b		
	<=			0b	0b	0b		
1.100.254.169	=>	all-systems.mcast.net		0b	0b	7b		
	<=			0b	0b	0b		
TX:	cum:	561KB	peak:	1.24Mb	rates:	1.21Kb	30.3Kb	99.2Kb
RX:		425KB		974Kb		1.86Kb	22.3Kb	71.7Kb
TOTAL:		986KB		1.31Mb		3.07Kb	52.6Kb	171Kb

4. Network Troubleshooting

```
(base) zianw@wangziyandeMacBook-Pro ~ % ping google.com
```

```
PING google.com (142.250.72.206): 56 data bytes
64 bytes from 142.250.72.206: icmp_seq=0 ttl=116 time=26.782 ms
64 bytes from 142.250.72.206: icmp_seq=1 ttl=116 time=27.135 ms
64 bytes from 142.250.72.206: icmp_seq=2 ttl=116 time=27.917 ms
64 bytes from 142.250.72.206: icmp_seq=3 ttl=116 time=46.173 ms
64 bytes from 142.250.72.206: icmp_seq=4 ttl=116 time=41.701 ms
64 bytes from 142.250.72.206: icmp_seq=5 ttl=116 time=32.436 ms
64 bytes from 142.250.72.206: icmp_seq=6 ttl=116 time=26.445 ms
64 bytes from 142.250.72.206: icmp_seq=7 ttl=116 time=28.952 ms
64 bytes from 142.250.72.206: icmp_seq=8 ttl=116 time=36.241 ms
64 bytes from 142.250.72.206: icmp_seq=9 ttl=116 time=25.033 ms
64 bytes from 142.250.72.206: icmp_seq=10 ttl=116 time=32.323 ms
64 bytes from 142.250.72.206: icmp_seq=11 ttl=116 time=31.155 ms
64 bytes from 142.250.72.206: icmp_seq=12 ttl=116 time=33.139 ms
64 bytes from 142.250.72.206: icmp_seq=13 ttl=116 time=73.871 ms
^C
```

```
--- google.com ping statistics ---
```

```
14 packets transmitted, 14 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 25.033/34.950/73.871/12.256 ms
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % traceroute google.com
traceroute to google.com (142.250.72.206), 64 hops max, 40 byte packets
```

```
 1  10.0.0.1 (10.0.0.1)  10.750 ms  6.283 ms  9.057 ms
 2  100.92.203.3 (100.92.203.3)  24.282 ms
   100.92.203.2 (100.92.203.2)  22.099 ms
   100.92.203.3 (100.92.203.3)  25.529 ms
 3  po-325-351-rur201.san jose.ca.sfba.comcast.net (162.151.30.209)  26.208 ms * 24.897 ms
 4  po-325-351-rur201.san jose.ca.sfba.comcast.net (162.151.30.209)  22.965 ms
   po-200-xar01.san jose.ca.sfba.comcast.net (68.85.86.173)  23.464 ms
   po-325-351-rur201.san jose.ca.sfba.comcast.net (162.151.30.209)  22.924 ms
 5  ae-199-rar01.santa clara.ca.sfba.comcast.net (68.87.226.109)  27.187 ms
   po-200-xar01.san jose.ca.sfba.comcast.net (68.85.86.173)  22.194 ms
   ae-199-rar01.santa clara.ca.sfba.comcast.net (68.87.226.109)  21.999 ms
 6  ae-199-rar01.santa clara.ca.sfba.comcast.net (68.87.226.109)  22.644 ms
   be-299-ar01.santa clara.ca.sfba.comcast.net (68.86.143.93)  26.431 ms
   ae-199-rar01.santa clara.ca.sfba.comcast.net (68.87.226.109)  53.405 ms
..
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % nmap google.com
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-01-30 11:39 PST
Nmap scan report for google.com (142.250.72.206)
Host is up (0.023s latency).
Other addresses for google.com (not scanned): 2607:f8b0:4005:813::200e
rDNS record for 142.250.72.206: sfo03s21-in-f14.1e100.net
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
```

```
Nmap done: 1 IP address (1 host up) scanned in 44.53 seconds
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % nslookup google.com
```

```
Server:      2001:558:feed::1
Address:     2001:558:feed::1#53
```

```
Non-authoritative answer:
```

```
Name:   google.com
Address: 142.250.72.206
```

(base) zianw@wangziyandeMacBook-Pro ~ % `ss -tln`

tcp6	0	0	*.50855	*. *	LISTEN
tcp4	0	0	*.50855	*. *	LISTEN
tcp6	0	0	::1.42050	*. *	LISTEN
tcp4	0	0	127.0.0.1.631	*. *	LISTEN
tcp6	0	0	::1.631	*. *	LISTEN
tcp6	0	0	*.5000	*. *	LISTEN
tcp4	0	0	*.5000	*. *	LISTEN
tcp6	0	0	*.7000	*. *	LISTEN
tcp4	0	0	*.7000	*. *	LISTEN
tcp4	0	0	*.88	*. *	LISTEN
tcp6	0	0	*.88	*. *	LISTEN
tcp4	0	0	*.5900	*. *	LISTEN
tcp6	0	0	*.5900	*. *	LISTEN

[illegible]

COMMAND	PID	USER	FD	TYPE	DEVICE	SIZE/OFF	NODE	NAME
loginwind	378	ziyanw	4u	IPv4	0x79d5ba7a19415a2f	0t0	UDP	*:*
rapportd	640	ziyanw	12u	IPv4	0x79d5ba83b13df33f	0t0	TCP	*:50855 (LISTEN)
rapportd	640	ziyanw	13u	IPv4	0x79d5ba7a1ad00e2f	0t0	UDP	*:*
rapportd	640	ziyanw	14u	IPv4	0x79d5ba7a1acf962f	0t0	UDP	*:*
rapportd	640	ziyanw	16u	IPv6	0x79d5ba888782c56f	0t0	TCP	*:50855 (LISTEN)
Google	650	ziyanw	112u	IPv4	0x79d5ba7a19423e2f	0t0	UDP	*:*
identitys	656	ziyanw	18u	IPv4	0x79d5ba7a1940d62f	0t0	UDP	*:*
identitys	656	ziyanw	23u	IPv4	0x79d5ba7a1940c22f	0t0	UDP	*:*
sharingd	679	ziyanw	4u	IPv4	0x79d5ba7a19cfbe2f	0t0	UDP	*:*
sharingd	679	ziyanw	5u	IPv4	0x79d5ba7a19cfb62f	0t0	UDP	*:*
sharingd	679	ziyanw	6u	IPv4	0x79d5ba7a19cfb22f	0t0	UDP	*:*
sharingd	679	ziyanw	7u	IPv4	0x79d5ba7a1941de2f	0t0	UDP	*:*
sharingd	679	ziyanw	10u	IPv4	0x79d5ba7a1941622f	0t0	UDP	*:*
sharingd	679	ziyanw	11u	IPv4	0x79d5ba7a193eae2f	0t0	UDP	*:*
sharingd	679	ziyanw	12u	IPv4	0x79d5ba7a193ed22f	0t0	UDP	*:*
sharingd	679	ziyanw	17u	IPv6	0x79d5ba7a1940de2f	0t0	UDP	*:50847
ControlCe	688	ziyanw	5u	IPv4	0x79d5ba83b13db9bf	0t0	TCP	*:7000 (LISTEN)
ControlCe	688	ziyanw	6u	IPv6	0x79d5ba8885fb056f	0t0	TCP	*:7000 (LISTEN)
ControlCe	688	ziyanw	7u	IPv4	0x79d5ba83b13dd0bf	0t0	TCP	*:5000 (LISTEN)
ControlCe	688	ziyanw	8u	IPv6	0x79d5ba8885fb0d6f	0t0	TCP	*:5000 (LISTEN)
ControlCe	688	ziyanw	18u	IPv4	0x79d5ba7a1942462f	0t0	UDP	*:*
WiFiAgent	719	ziyanw	3u	IPv4	0x79d5ba7a1942362f	0t0	UDP	*:*
Google	939	ziyanw	19u	IPv6	0x79d5ba8885f8956f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:51986->[2001:558:feed:443::148]:443 (ESTABLISHED)
Google	939	ziyanw	20u	IPv6	0x79d5ba8885f8056f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:54189->[2606:4700:4400::ac40:9bd1]:443 (ESTABLISHED)
Google	939	ziyanw	22u	IPv6	0x79d5ba8885f8756f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:52063->[2602:f79a::2]:443 (ESTABLISHED)
Google	939	ziyanw	25u	IPv6	0x79d5ba8885f88d6f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:51987->[2001:558:feed:443::147]:443 (ESTABLISHED)
Google	939	ziyanw	26u	IPv6	0x79d5ba8885f7ed6f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:54197->[2606:4700:4400::ac40:9bd1]:443 (ESTABLISHED)
Google	939	ziyanw	29u	IPv4	0x79d5ba83b13dfefbf	0t0	TCP	10.0.0.13:54198->34.233.184.229:443 (ESTABLISHED)
Google	939	ziyanw	41u	IPv4	0x79d5ba83b13da2bf	0t0	TCP	10.0.0.13:52045->54.83.37.232:443 (ESTABLISHED)
Google	939	ziyanw	44u	IPv6	0x79d5ba8885f7fd6f	0t0	TCP	[2601:646:a180:5db0:4df4:e8f3:dab4:2e34]:51411->[2607:f8b0:4023:c06::bc]:5228 (ESTABLISHED)
Google	939	ziyanw	169u	IPv6	0x79d5ba7a19cfce2f	0t0	UDP	*:5353
Google	939	ziyanw	170u	IPv4	0x79d5ba7a19cfd22f	0t0	UDP	*:5353
Google	939	ziyanw	171u	IPv6	0x79d5ba7a19cfd62f	0t0	UDP	*:5353
Google	939	ziyanw	172u	IPv6	0x79d5ba7a19cf9e2f	0t0	UDP	*:5353
Google	939	ziyanw	173u	IPv6	0x79d5ba7a19cfaa2f	0t0	UDP	*:5353
Google	939	ziyanw	174u	IPv6	0x79d5ba7a19cfae2f	0t0	UDP	*:5353
Google	939	ziyanw	175u	IPv6	0x79d5ba7a19cfba2f	0t0	UDP	*:5353
Google	939	ziyanw	176u	IPv6	0x79d5ba7a19cfc22f	0t0	UDP	*:5353
Google	939	ziyanw	177u	IPv6	0x79d5ba7a19cfca2f	0t0	UDP	*:5353
Microsoft	22275	ziyanw	14u	IPv6	0x79d5ba8887829d6f	0t0	TCP	[::1]:42050 (LISTEN)
WeChat	44205	ziyanw	5u	IPv4	0x79d5ba7a1940562f	0t0	UDP	*:*
WeChat	44205	ziyanw	14u	IPv4	0x79d5ba7a1940422f	0t0	UDP	*:*
WeChat	44205	ziyanw	75u	IPv6	0x79d5ba8885f85d6f	0t0	TCP	10.0.0.13:51521->43.130.30.185:443 (ESTABLISHED)
WeChat	44205	ziyanw	94u	IPv6	0x79d5ba8885f8856f	0t0	TCP	10.0.0.13:51516->43.130.30.185:80 (CLOSE_WAIT)
chronod	71970	ziyanw	14u	IPv6	0x79d5ba7a19cf1e2f	0t0	UDP	*:51746
assistant	98849	ziyanw	25u	IPv4	0x79d5ba7a193eaa2f	0t0	UDP	*:*

5. Advanced Packet Analysis

```
(base) zianw@wangziyandeMacBook-Pro ~ % sudo tcpdump -i en0 -w capture.pcap
```

Password:

```
tcpdump: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
^C2694 packets captured
2697 packets received by filter
0 packets dropped by kernel
```

```
(base) zianw@wangziyandeMacBook-Pro ~ % tcpdump -r capture.pcap
reading from file capture.pcap, link-type EN10MB (Ethernet)
12:01:51.356700 IP 10.0.0.142.62525 > broadcasthost.6667: UDP, length 204
12:01:52.585433 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
12:01:53.814276 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
12:01:54.735812 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
12:01:54.841013 IP 169.254.100.1.50760 > 239.255.255.250.ssd: UDP, length 191
12:01:54.841013 IP 169.254.100.1.50760 > 239.255.255.250.ssd: UDP, length 195
12:01:54.841014 IP 169.254.100.1.50760 > 239.255.255.250.ssd: UDP, length 199
12:01:54.841014 IP 169.254.100.1.50759 > 239.255.255.250.ssd: UDP, length 192
12:01:55.000734 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.54201 > 2602:f79a::1.https: Flags [.), a
ck 1560712645, win 2048, length 0
12:01:55.030544 IP6 2602:f79a::1.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.54201: Flags [.), a
ck 1, win 16, options [nop,nop,TS val 3033604669 ecr 3062809761], length 0
12:01:55.657503 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
12:01:55.863729 IP6 fe80::8aef:16ff:fecc:8792 > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34: ICMP6, neig
hbor solicitation, who has 2601:646:a180:5db0:4df4:e8f3:dab4:2e34, length 32
12:01:55.863824 IP6 wangziyandemacbook-pro.local > fe80::8aef:16ff:fecc:8792: ICMP6, neighbor adver
tisement, tgt is 2601:646:a180:5db0:4df4:e8f3:dab4:2e34, length 24
12:01:56.271914 IP 10.0.0.142.62525 > broadcasthost.6667: UDP, length 204
12:01:56.579104 IP6 fe80::8aef:16ff:fecc:8792 > ff02::1: ICMP6, router advertisement, length 120
12:01:57.370010 IP 10.0.0.13.54210 > ec2-52-2-210-102.compute-1.amazonaws.com.https: Flags [P.], se
q 1346468069:1346469470, ack 2453869913, win 2048, options [nop,nop,TS val 1647457196 ecr 196180564
6], length 1401
12:01:57.370052 IP 10.0.0.13.54210 > ec2-52-2-210-102.compute-1.amazonaws.com.https: Flags [P.], se
q 1401:1447, ack 1, win 2048, options [nop,nop,TS val 1647457196 ecr 1961805646], length 46
12:01:57.370129 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [P.], seq 3135146795:3135146992, ack 524204114, win 2048, options [nop,nop,TS val 3131339093 ec
r 2040403034], length 197
12:01:57.370142 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [P.], seq 197:243, ack 1, win 2048, options [nop,nop,TS val 3131339093 ecr 2040403034], length
46
12:01:57.370157 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [P.], seq 243:440, ack 1, win 2048, options [nop,nop,TS val 3131339093 ecr 2040403034], length
197
12:01:57.370170 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [P.], seq 440:637, ack 1, win 2048, options [nop,nop,TS val 3131339093 ecr 2040403034], length
197
12:01:57.417763 IP6 2001:558:feed:443::148.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986: Fl
ags [P.], seq 1:46, ack 197, win 51120, options [nop,nop,TS val 2040411018 ecr 3131339093], length
45
12:01:57.417901 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [.), ack 46, win 2047, options [nop,nop,TS val 3131339141 ecr 2040411018], length 0
12:01:57.419349 IP6 2001:558:feed:443::148.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986: Fl
ags [P.], seq 214:260, ack 637, win 51120, options [nop,nop,TS val 2040411019 ecr 3131339093], leng
th 46
12:01:57.419350 IP6 2001:558:feed:443::148.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986: Fl
ags [.), ack 440, win 51059, options [nop,nop,TS val 2040411018 ecr 3131339093], length 0
12:01:57.419350 IP6 2001:558:feed:443::148.https > 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986: Fl
ags [P.], seq 46:214, ack 637, win 51120, options [nop,nop,TS val 2040411018 ecr 3131339093], lengt
h 168
12:01:57.419688 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
ags [.), ack 260, win 2044, options [nop,nop,TS val 3131339143 ecr 2040411018], length 0
12:01:57.419688 IP6 2601:646:a180:5db0:4df4:e8f3:dab4:2e34.51986 > 2001:558:feed:443::148.https: Fl
```